

Diagnostic Contribution of the VWF:CB Assay in von Willebrand Disease: A Pilot Study

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BACKGROUND

Von Willebrand disease (VWD) is the most common inherited bleeding disorder. Its diagnosis requires both functional and antigenic assessment of von Willebrand factor (VWF). The VWF collagen-binding assay (VWF:CB) evaluates the ability of VWF to bind collagen and may provide additional diagnostic value, although it remains underused in routine practice.

OBJECTIVE

- To evaluate the contribution of VWF: CB assay to the biological diagnosis of VWD by comparing its performance with established hemostatic parameters.
- To distinguish type 2A from type 2M of Von Willebrand disease.

MATERIALS & METHODS

STUDY PERIOD

November 2024 May 2025

STUDY POPULATION: 19 patients VS Control group.

PATIENT REFERRAL

Hematology Département _CHUO

Pediatrics Département _EHS Canastel,Oran

PATIENT CHARACTERISTICS

Sex distribution 10 female | 9 male

Sex-ratio(M/F):0,9

Median age =13,47 years

VWF:CB ASSAY

Reagent: ASSERACHROM VWF:CB by ELISA

COMPARATIVE PARAMETERS

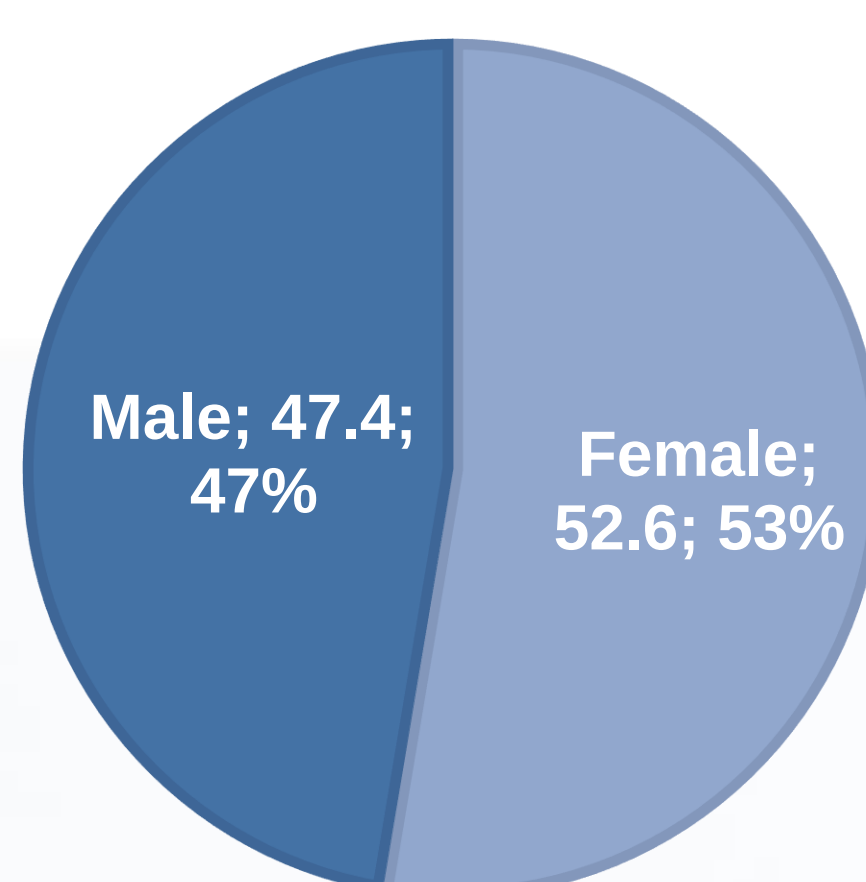
VWF:CB results were compared with :

- VWF:Ag, Factor VIII levels , Control values,Sex and ABO blood group.

STATISTICAL ANALYSIS

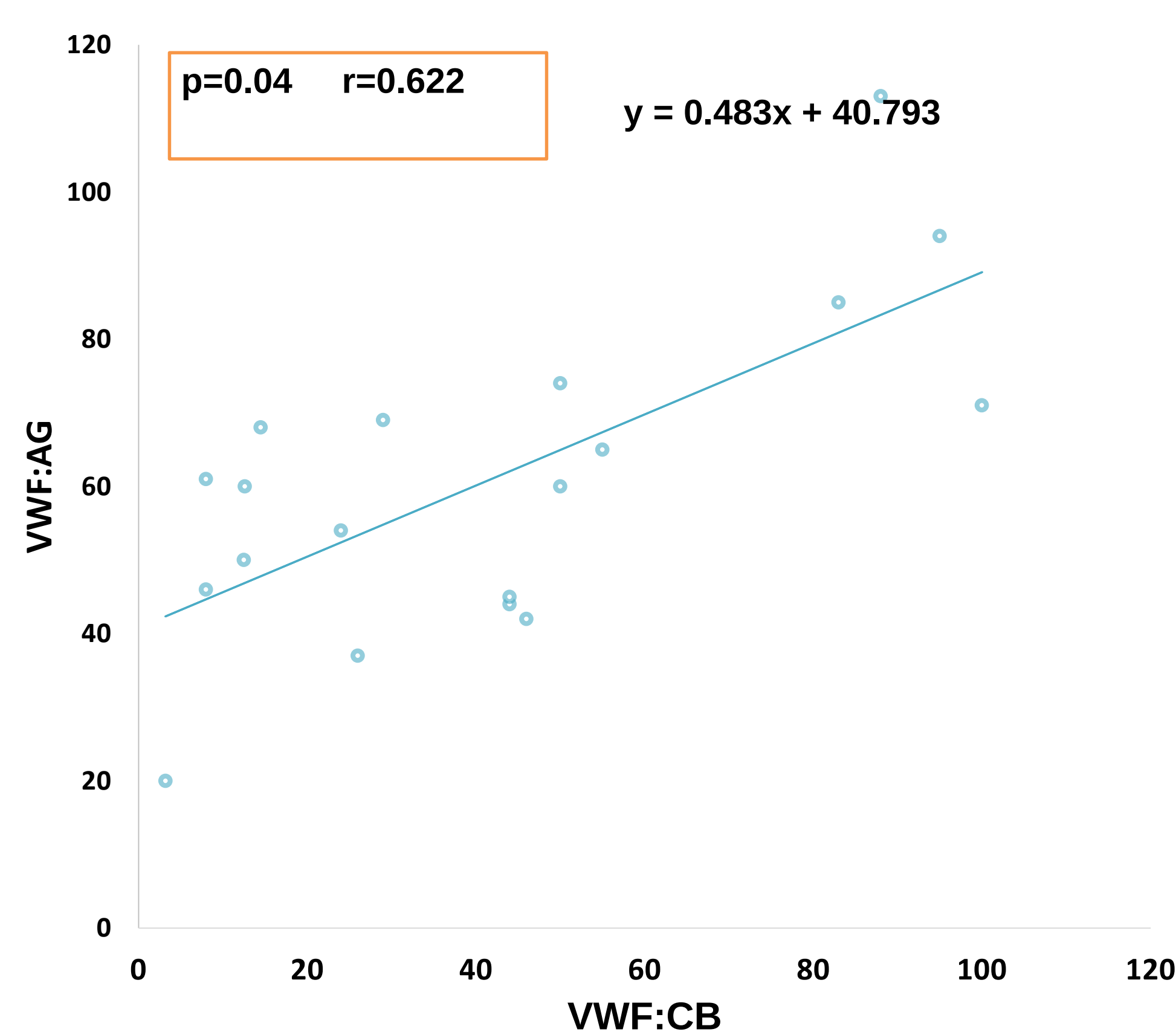
Correlation and group comparison by IBM SPSS Statistics-27

■ Female ■ Male



GRAPH 1 : Sex distribution

GRAPH 2 : Correlation between VWF:CB vs VWF:Ag levels

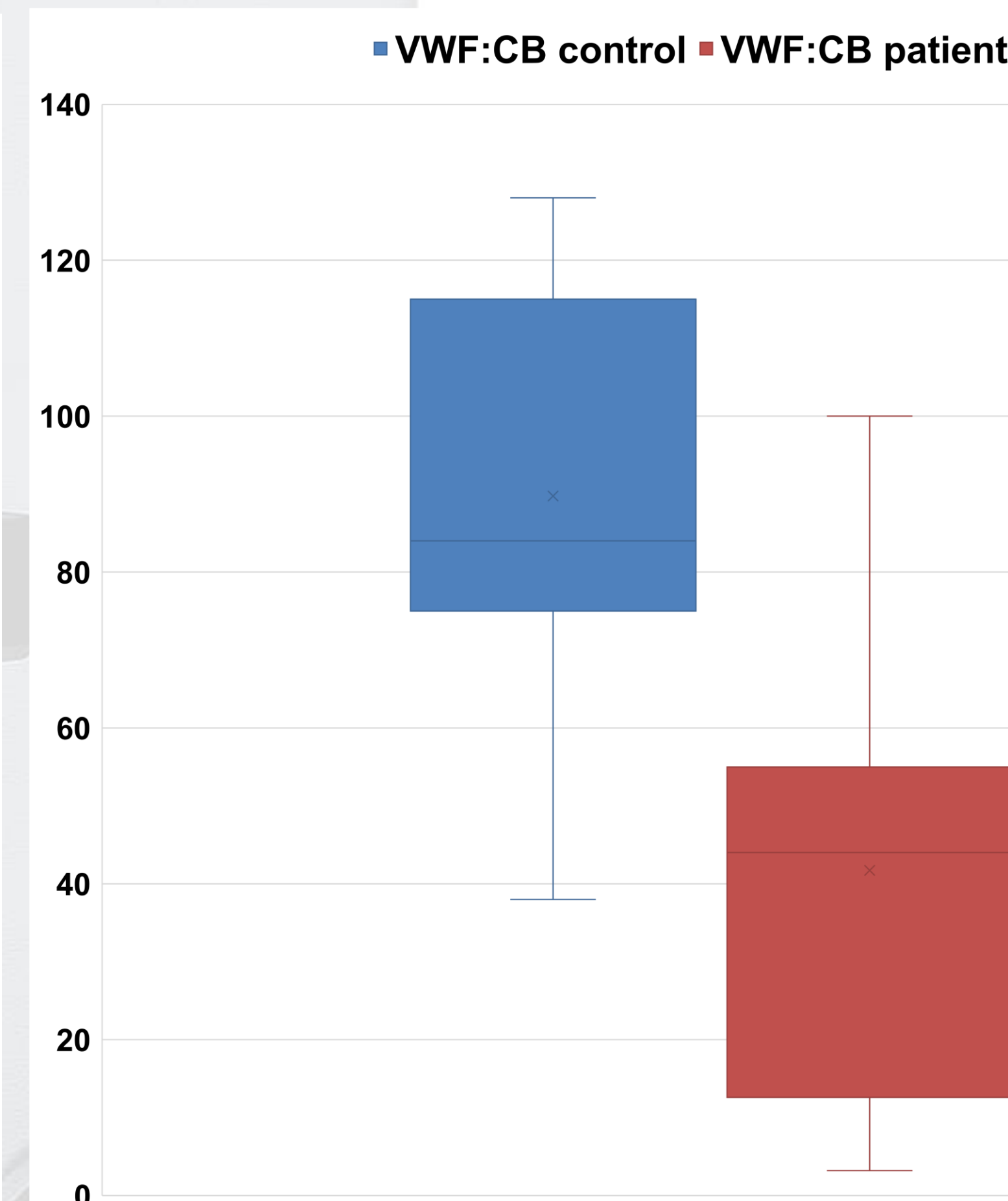


$\rho = 0.622$ | $p = 0.04$ significant correlation
Kendall's Tau-B: $p = 0.007$; $r = 0.456$

❖ Other findings:

- ✓ No significant differences were observed according to sex or ABO blood group
- ✓ No significant relationship was found between VWF:CB and FVIII.

GRAPH 3 :Comparison of VWF:CB between Patients and Controls.



VWF:CB lower in patients
 $r = 0.875$

CONCLUSION

- VWF:CB appears to be a useful complementary assay for the biological evaluation of von Willebrand disease, with potential discriminatory value between patients and controls. Further studies with larger cohorts and rigorous assay standardization are needed before routine implementation.

REFERENCES

- [1] Gothot A, Lecut C, Péters P, Vandenbosch K. Maladie de von Willebrand :corrélation entre le test Collagène Binding (VWF:CB) et le dosage antigénique de facteur von Willebrand (VWF:Ag) 2024
- [2] Kirkeby MH et al. "Intraindividual variability of von Willebrand factor and the impact of sex, age and blood group on factor levels." Blood Coagulation & Fibrinolysis. 2025 May
- [3] Flood VH, Gill JC, Friedman KD, et al; Zimmerman Program Investigators. Collagen binding provides a sensitive screen for variant von Willebrand disease. Clin Chem. 2013;59(4):684-691.

DISCUSSION

- ✓ The significant association between VWF:CB and VWF:Ag supports the complementary role of VWF:CB in the biological assessment of VWD. [1]
- ✓ The absence of significant differences according to sex or ABO blood group suggests limited influence of these variables [2], while the lack of association with FVIII supports the functional specificity of VWF:CB.
- ✓ The lower VWF:CB values observed in patients compared with control. highlight its potential discriminatory capacity. [3]